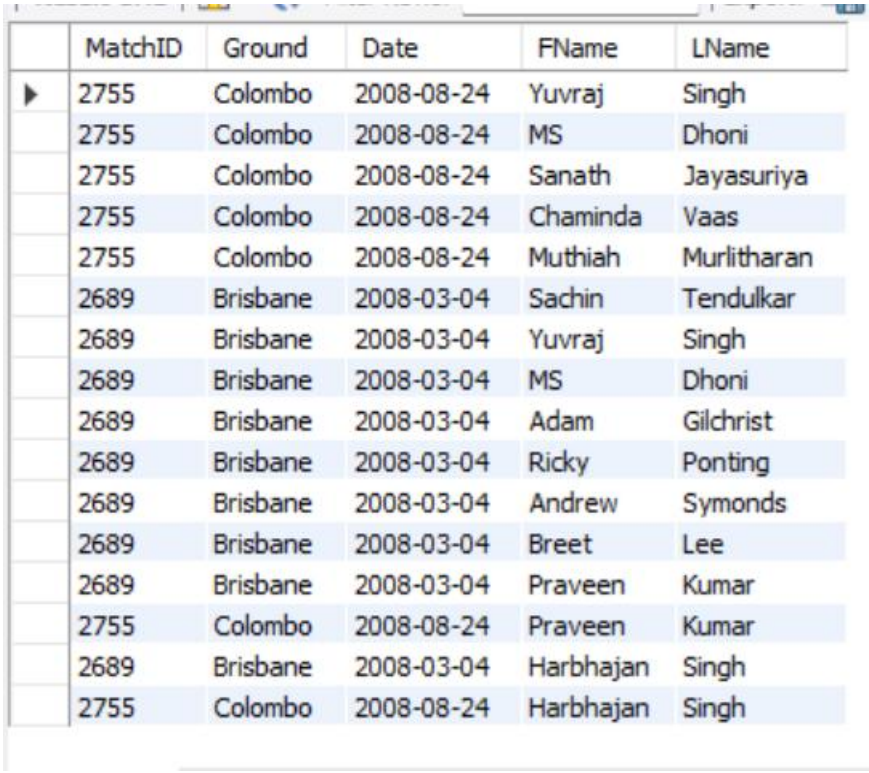
LAB 6: Implementation of Various Join Operations

Aim: *Understanding and applying INNER JOIN, LEFT OUTER JOIN, RIGHT OUTER JOIN, FULL OUTER JOIN, SELF JOIN, CROSS JOIN, and Multi-table Joins on relational schemas.*

1 [Inner Join]: Write an SQL query using INNER JOIN to display the MatchID, Ground, Date, and the Player's full name (FName, LName) for every player who batted in each match.

Select o.MatchID,o.Ground,o.Date,p.FName,p.LName from odimatch o inner join batting b on o.MatchID = b.MatchID inner join player p on b.PID = p.PlayerID;



MatchID	Ground	Date	FName	LName
2755	Colombo	2008-08-24	Yuvraj	Singh
2755	Colombo	2008-08-24	MS	Dhoni
2755	Colombo	2008-08-24	Sanath	Jayasuriya
2755	Colombo	2008-08-24	Chaminda	Vaas
2755	Colombo	2008-08-24	Muthiah	Murlitharan
2689	Brisbane	2008-03-04	Sachin	Tendulkar
2689	Brisbane	2008-03-04	Yuvraj	Singh
2689	Brisbane	2008-03-04	MS	Dhoni
2689	Brisbane	2008-03-04	Adam	Gilchrist
2689	Brisbane	2008-03-04	Ricky	Ponting
2689	Brisbane	2008-03-04	Andrew	Symonds
2689	Brisbane	2008-03-04	Breet	Lee
2689	Brisbane	2008-03-04	Praveen	Kumar
2755	Colombo	2008-08-24	Praveen	Kumar
2689	Brisbane	2008-03-04	Harbhajan	Singh
2755	Colombo	2008-08-24	Harbhajan	Singh

2 [Inner Join with Filter]: Display Player full name, Country, NRuns, and Six for all batting performances where the player scored at least one six (Six > 0).

select p.Fname,p.LName,p.Country,b.NRuns,b.Six from player p inner join batting b on p.PlayerID = b.PID;

Fname	LName	Country	NRuns	Six
Yuvraj	Singh	India	0	0
MS	Dhoni	India	71	0
Sanath	Jayasuriya	Sri Lanka	60	2
Chaminda	Vaas	Sri Lanka	17	0
Muthiah	Murlitharan	Sri Lanka	1	0
Sachin	Tendulkar	India	91	0
Yuvraj	Singh	India	38	2
MS	Dhoni	India	36	1
Adam	Gilchrist	Australia	2	0
Ricky	Ponting	Australia	1	0
Andrew	Symonds	Australia	42	1
Breet	Lee	Australia	7	0
Praveen	Kumar	India	7	0
Praveen	Kumar	India	2	0
Harbhajan	Singh	India	3	0
Harbhajan	Singh	India	2	0

3 [Left Outer Join]: Write a query using LEFT OUTER JOIN to list all players (PlayerID, FName, LName) along with their batting scores (MatchID, NRuns). Ensure players who have never batted in any match are also displayed with NULL values.

select p.PlayerID,p.FName,p.LName,b.MatchID,b.NRuns from player p left outer join batting b on p.PlayerID = b.PID;

PlayerID	FName	LName	MatchID	NRuns
21001	Virender	Sehwag	NULL	NULL
23001	Yuvraj	Singh	2755	0
23001	Yuvraj	Singh	2689	38
24001	Andrew	Symonds	2689	42
25001	MS	Dhoni	2755	71
25001	MS	Dhoni	2689	36
27001	Praveen	Kumar	2689	7
27001	Praveen	Kumar	2755	2
27002	Ishant	Sharma	NULL	NULL
89001	Sachin	Tendulkar	2689	91
90001	Brain	Lara	NULL	NULL
90002	Anil	Kumble	NULL	NULL
91001	Sanath	Jayasuriya	2755	60
92001	Shane	Warne	NULL	NULL
92002	Muthiah	Murlitharan	2755	1
92003	Inzamam	UI-Huq	NULL	NULL
93002	Heath	Streak	NULL	NULL

PlayerID	FName	LName	MatchID	NRuns
93003	Gary	Kirsten	NULL	NULL
94002	Chaminda	Vaas	2755	17
94004	Stephen	Fleming	NULL	NULL
95001	Ricky	Ponting	2689	1
95002	Shaun	Pollock	NULL	NULL
95003	Jacques	Kallis	NULL	NULL
96001	Rahul	Dravid	NULL	NULL
96002	Herschelle	Gibbs	NULL	NULL
96003	Saurav	Ganguly	NULL	NULL
97004	Daniel	Vettori	NULL	NULL
98001	Shahid	Afridi	NULL	NULL
98002	Harbha...	Singh	2689	3
98002	Harbha...	Singh	2755	2
99001	Breet	Lee	2689	7
99002	Adam	Gilchrist	2689	2
99003	Michael	Vaughan	NULL	NULL
300311	Michael	NULL	NULL	NULL

4 [Right Outer Join]: Retrieve all bowling performance entries (MatchID, NOvers, Nwickets) along with player details (FName, LName, Country) using a RIGHT OUTER JOIN, showing all bowling entries even if player profiles are missing.

MatchID	NOvers	Nwickets	FName	LName	Country
NULL	NULL	NULL	Virender	Sehwag	India
2689	3	0	Yuvraj	Singh	India
2755	10	2	Yuvraj	Singh	India
2689	3	1	Andrew	Symonds	Australia
NULL	NULL	NULL	MS	Dhoni	India
NULL	NULL	NULL	Praveen	Kumar	India
NULL	NULL	NULL	Ishant	Sharma	India
NULL	NULL	NULL	Sachin	Tendulkar	India
NULL	NULL	NULL	Brain	Lara	West Indies
NULL	NULL	NULL	Anil	Kumble	India
2755	4	0	Sanath	Jayasuriya	Sri Lanka
NULL	NULL	NULL	Shane	Warne	Australia
2755	10	1	Muthiah	Murlitharan	Sri Lanka
NULL	NULL	NULL	Inzamam	UI-Huq	Pakistan
NULL	NULL	NULL	Heath	Streak	Zimbabwe
NULL	NULL	NULL	Gary	Kirsten	South Africa
2755	9	1	Chaminda	Vaas	Sri Lanka

NULL	NULL	NULL	Stephen	Fleming	New Zealand
NULL	NULL	NULL	Ricky	Ponting	Australia
NULL	NULL	NULL	Shaun	Pollock	South Africa
NULL	NULL	NULL	Jacques	Kallis	South Africa
NULL	NULL	NULL	Rahul	Dravid	India
NULL	NULL	NULL	Herschelle	Gibbs	South Africa
NULL	NULL	NULL	Saurav	Ganguly	India
NULL	NULL	NULL	Daniel	Vettori	New Zealand
NULL	NULL	NULL	Shahid	Afridi	Pakistan
2755	10	3	Harbha...	Singh	India
2689	10	1	Harbha...	Singh	India
2689	10	1	Breet	Lee	Australia
NULL	NULL	NULL	Adam	Gilchrist	Australia
NULL	NULL	NULL	Michael	Vaughan	England
NULL	NULL	NULL	Michael	NULL	NULL

5 [Full Outer Join / Union]: Write an SQL query to perform a FULL OUTER JOIN (or simulate using LEFT JOIN UNION RIGHT JOIN) between Batting and Bowling tables on MatchID and PID to view all combined match participation records.

Fours	FOW	HOut	MatchID	Mts	Nballs	NRuns	PID	POrder	Six	Bowling_MatchID	Bowling_PID	NOvers	Maidens	Bowling_Nruns	Nwickets
0	51	C	2755	12	6	0	23001	3	0	2755	23001	10	0	53	2
4	232	C	2755	104	80	71	25001	5	0	NULL	NULL	NULL	NULL	NULL	NULL
8	74	C	2755	85	52	60	91001	1	2	2755	91001	4	0	29	0
1	157	LBW	2755	23	29	17	94002	7	0	2755	94002	9	1	40	1
0	NO	NO	2755	11	7	1	92002	11	0	2755	92002	10	0	56	1
7	205	C	2689	176	121	91	89001	2	0	NULL	NULL	NULL	NULL	NULL	NULL
2	175	C	2689	27	38	38	23001	4	2	2689	23001	3	0	15	0
2	240	C	2689	52	37	36	25001	5	1	NULL	NULL	NULL	NULL	NULL	NULL
0	2	C	2689	3	3	2	99002	1	0	NULL	NULL	NULL	NULL	NULL	NULL
0	8	C	2689	9	7	1	95001	3	0	NULL	NULL	NULL	NULL	NULL	NULL
2	123	C	2689	81	56	42	24001	5	1	2689	24001	3	0	27	1
0	228	B	2689	20	12	7	99001	8	0	2689	99001	10	0	58	1
1	255	C	2689	7	7	7	27001	9	0	NULL	NULL	NULL	NULL	NULL	NULL
0	257	B	2755	7	6	2	27001	9	0	NULL	NULL	NULL	NULL	NULL	NULL
0	249	LBW	2689	7	3	3	98002	8	0	2689	98002	10	0	44	1
0	253	RO	2755	7	4	2	98002	8	0	2755	98002	10	0	40	3

6 [3-Table Inner Join]: List Player.FName, Player.LName, ODIMatch.Ground, ODIMatch.Date, and Batting.NRuns for all matches played where 'India' was either Team1 or Team2.

select p.FName,p.LName,o.Ground,o.Date,b.NRuns from player p join batting b on p.PlayerID = b.PID join odimatch o on b.MatchID=o.MatchID where o.Team1='India' or o.Team2='India';

FName	LName	Ground	Date	NRuns
Yuvraj	Singh	Colombo	2008-08-24	0
MS	Dhoni	Colombo	2008-08-24	71
Sanath	Jayasuriya	Colombo	2008-08-24	60
Chaminda	Vaas	Colombo	2008-08-24	17
Muthiah	Murlitharan	Colombo	2008-08-24	1
Sachin	Tendulkar	Brisbane	2008-03-04	91
Yuvraj	Singh	Brisbane	2008-03-04	38
MS	Dhoni	Brisbane	2008-03-04	36
Adam	Gilchrist	Brisbane	2008-03-04	2
Ricky	Ponting	Brisbane	2008-03-04	1
Andrew	Symonds	Brisbane	2008-03-04	42
Breet	Lee	Brisbane	2008-03-04	7
Praveen	Kumar	Brisbane	2008-03-04	7
Praveen	Kumar	Colombo	2008-08-24	2
Harbhajan	Singh	Brisbane	2008-03-04	3
Harbhajan	Singh	Colombo	2008-08-24	2

7 [4-Table Multi-Join]: Display player full name, Ground, Date, Batting.NRuns, and Bowling.Nwickets for all players who both BATTED and BOWLED in the exact same match.

select p.FName,p.LName,o.Ground,o.Date,b.NRuns,bow.Nwickets from player p join batting b on p.PlayerID=b.PID join bowling bow on b.PID = bow.PID join odimatch o on bow.MatchID = o.MatchID;

FName	LName	Ground	Date	NRuns	Nwickets
Breet	Lee	Brisbane	2008-03-04	7	1
Andrew	Symonds	Brisbane	2008-03-04	42	1
Yuvraj	Singh	Brisbane	2008-03-04	0	0
Yuvraj	Singh	Brisbane	2008-03-04	38	0
Chaminda	Vaas	Colombo	2008-08-24	17	1
Muthiah	Murlitharan	Colombo	2008-08-24	1	1
Sanath	Jayasuriya	Colombo	2008-08-24	60	0
Yuvraj	Singh	Colombo	2008-08-24	0	2
Yuvraj	Singh	Colombo	2008-08-24	38	2
Harbhajan	Singh	Colombo	2008-08-24	3	3
Harbhajan	Singh	Colombo	2008-08-24	2	3
Harbhajan	Singh	Brisbane	2008-03-04	3	1
Harbhajan	Singh	Brisbane	2008-03-04	2	1

8 [Self Join - Attribute Match]: Write a query using a SELF JOIN on the Player table to find pairs of players who were born in the same birth place (Bplace). Display Player 1 Name, Player 2 Name, and Bplace (exclude self-pairings and duplicate reversed pairs).

select p1.FName, p1.LName, p2.FName,p2.LName,p1.Bplace from player p1 join player p2 on p1.Bplace=p2.Bplace and p1.PlayerID < p2.PlayerID;

FName	LName	FName	LName	Bplace
Virender	Sehwag	Ishant	Sharma	Delhi
Gary	Kirsten	Jacques	Kallis	Cape Town
Gary	Kirsten	Herschelle	Gibbs	Cape Town
Jacques	Kallis	Herschelle	Gibbs	Cape Town

9 [Self Join - Match Analysis]: Using a SELF JOIN on ODIMatch, find pairs of distinct matches that were played on the same Ground on different dates.

select o1.MatchID,o1.Ground,o1.Date,o2.MatchID,o2.Ground,o2.Date from odimatch o1 join odimatch o2 on o1.Ground=o2.Ground where o1.Date<>o2.Date and o1.MatchID<o2.MatchID;

MatchID	Ground	Date	MatchID	Ground	Date
2404	Colombo	2006-08-16	2750	Colombo	2008-08-24
2404	Colombo	2006-08-16	2755	Colombo	2008-08-24

10 [Cross Join]: Generate a Cartesian product (CROSS JOIN) between players from 'India' and all matches in ODIMatch to display all hypothetical player-match combinations.

select * from odimatch o cross join player p where p.Country = 'India';

MatchID	Team1	Team2	Ground	Date	Winner	PlayerID	LName	FName	Country	Yborn	Bplace	FTest
2324	Pakistan	India	Peshwar	2006-02-06	Team1	98002	Singh	Harbhajan	India	1980	Jalandhar	1998
2324	Pakistan	India	Peshwar	2006-02-06	Team1	96003	Ganguly	Saurav	India	1972	Calcutta	1996
2324	Pakistan	India	Peshwar	2006-02-06	Team1	96001	Dravid	Rahul	India	1973	Indore	1996
2324	Pakistan	India	Peshwar	2006-02-06	Team1	90002	Kumble	Anil	India	1970	Bangalore	1990
2324	Pakistan	India	Peshwar	2006-02-06	Team1	89001	Tendulkar	Sachin	India	1973	Mumbai	1989
2324	Pakistan	India	Peshwar	2006-02-06	Team1	27002	Sharma	Ishant	India	1988	Delhi	2007
2324	Pakistan	India	Peshwar	2006-02-06	Team1	27001	Kumar	Praveen	India	1986	Meerut	NULL
2324	Pakistan	India	Peshwar	2006-02-06	Team1	25001	Dhoni	MS	India	1981	Ranchi	2005
2324	Pakistan	India	Peshwar	2006-02-06	Team1	23001	Singh	Yuvraj	India	1981	Chandigarh	2003
2324	Pakistan	India	Peshwar	2006-02-06	Team1	21001	Sehwag	Virender	India	1978	Delhi	2001
2327	Pakistan	India	Rawalpi...	2006-02-11	Team2	98002	Singh	Harbhajan	India	1980	Jalandhar	1998
2327	Pakistan	India	Rawalpi...	2006-02-11	Team2	96003	Ganguly	Saurav	India	1972	Calcutta	1996
2327	Pakistan	India	Rawalpi...	2006-02-11	Team2	96001	Dravid	Rahul	India	1973	Indore	1996
2327	Pakistan	India	Rawalpi...	2006-02-11	Team2	90002	Kumble	Anil	India	1970	Bangalore	1990
2327	Pakistan	India	Rawalpi...	2006-02-11	Team2	89001	Tendulkar	Sachin	India	1973	Mumbai	1989
2327	Pakistan	India	Rawalpi...	2006-02-11	Team2	27002	Sharma	Ishant	India	1988	Delhi	2007
2327	Pakistan	India	Rawalpi...	2006-02-11	Team2	27001	Kumar	Praveen	India	1986	Meerut	NULL

LAKSHAY_32_D6EC_DBMS_LAB

MatchID	Team1	Team2	Ground	Date	Winner	PlayerID	LName	FName	Country	Yborn	Bplace	FTest
2327	Pakistan	India	Rawalpi...	2006-02-11	Team2	25001	Dhoni	MS	India	1981	Ranchi	2005
2327	Pakistan	India	Rawalpi...	2006-02-11	Team2	23001	Singh	Yuvraj	India	1981	Chandigarh	2003
2327	Pakistan	India	Rawalpi...	2006-02-11	Team2	21001	Sehwag	Virender	India	1978	Delhi	2001
2357	India	England	Delhi	2006-03-28	Team1	98002	Singh	Harbhajan	India	1980	Jalandhar	1998
2357	India	England	Delhi	2006-03-28	Team1	96003	Ganguly	Saurav	India	1972	Calcutta	1996
2357	India	England	Delhi	2006-03-28	Team1	96001	Dravid	Rahul	India	1973	Indore	1996
2357	India	England	Delhi	2006-03-28	Team1	90002	Kumble	Anil	India	1970	Bangalore	1990
2357	India	England	Delhi	2006-03-28	Team1	89001	Tendulkar	Sachin	India	1973	Mumbai	1989
2357	India	England	Delhi	2006-03-28	Team1	27002	Sharma	Ishant	India	1988	Delhi	2007
2357	India	England	Delhi	2006-03-28	Team1	27001	Kumar	Praveen	India	1986	Meerut	NULL
2357	India	England	Delhi	2006-03-28	Team1	25001	Dhoni	MS	India	1981	Ranchi	2005
2357	India	England	Delhi	2006-03-28	Team1	23001	Singh	Yuvraj	India	1981	Chandigarh	2003
2357	India	England	Delhi	2006-03-28	Team1	21001	Sehwag	Virender	India	1978	Delhi	2001
2377	West I...	India	Kingston	2006-05-18	Team2	98002	Singh	Harbhajan	India	1980	Jalandhar	1998
2377	West I...	India	Kingston	2006-05-18	Team2	96003	Ganguly	Saurav	India	1972	Calcutta	1996
2377	West I...	India	Kingston	2006-05-18	Team2	96001	Dravid	Rahul	India	1973	Indore	1996
2377	West I...	India	Kingston	2006-05-18	Team2	90002	Kumble	Anil	India	1970	Bangalore	1990

MatchID	Team1	Team2	Ground	Date	Winner	PlayerID	LName	FName	Country	Yborn	Bplace	FTest
2377	West I...	India	Kingston	2006-05-18	Team2	89001	Tendulkar	Sachin	India	1973	Mumbai	1989
2377	West I...	India	Kingston	2006-05-18	Team2	27002	Sharma	Ishant	India	1988	Delhi	2007
2377	West I...	India	Kingston	2006-05-18	Team2	27001	Kumar	Praveen	India	1986	Meerut	NULL
2377	West I...	India	Kingston	2006-05-18	Team2	25001	Dhoni	MS	India	1981	Ranchi	2005
2377	West I...	India	Kingston	2006-05-18	Team2	23001	Singh	Yuvraj	India	1981	Chandigarh	2003
2377	West I...	India	Kingston	2006-05-18	Team2	21001	Sehwag	Virender	India	1978	Delhi	2001
2404	Sri Lanka	India	Colombo	2006-08-16	Aband...	98002	Singh	Harbhajan	India	1980	Jalandhar	1998
2404	Sri Lanka	India	Colombo	2006-08-16	Aband...	96003	Ganguly	Saurav	India	1972	Calcutta	1996
2404	Sri Lanka	India	Colombo	2006-08-16	Aband...	96001	Dravid	Rahul	India	1973	Indore	1996
2404	Sri Lanka	India	Colombo	2006-08-16	Aband...	90002	Kumble	Anil	India	1970	Bangalore	1990
2404	Sri Lanka	India	Colombo	2006-08-16	Aband...	89001	Tendulkar	Sachin	India	1973	Mumbai	1989
2404	Sri Lanka	India	Colombo	2006-08-16	Aband...	27002	Sharma	Ishant	India	1988	Delhi	2007
2404	Sri Lanka	India	Colombo	2006-08-16	Aband...	27001	Kumar	Praveen	India	1986	Meerut	NULL
2404	Sri Lanka	India	Colombo	2006-08-16	Aband...	25001	Dhoni	MS	India	1981	Ranchi	2005
2404	Sri Lanka	India	Colombo	2006-08-16	Aband...	23001	Singh	Yuvraj	India	1981	Chandigarh	2003
2404	Sri Lanka	India	Colombo	2006-08-16	Aband...	21001	Sehwag	Virender	India	1978	Delhi	2001
2440	India	Austr...	Mohali	2006-10-29	Team2	98002	Singh	Harbhajan	India	1980	Jalandhar	1998

MatchID	Team1	Team2	Ground	Date	Winner	PlayerID	LName	FName	Country	Yborn	Bplace	FTest
2440	India	Austr...	Mohali	2006-10-29	Team2	96003	Ganguly	Saurav	India	1972	Calcutta	1996
2440	India	Austr...	Mohali	2006-10-29	Team2	96001	Dravid	Rahul	India	1973	Indore	1996
2440	India	Austr...	Mohali	2006-10-29	Team2	90002	Kumble	Anil	India	1970	Bangalore	1990
2440	India	Austr...	Mohali	2006-10-29	Team2	89001	Tendulkar	Sachin	India	1973	Mumbai	1989
2440	India	Austr...	Mohali	2006-10-29	Team2	27002	Sharma	Ishant	India	1988	Delhi	2007
2440	India	Austr...	Mohali	2006-10-29	Team2	27001	Kumar	Praveen	India	1986	Meerut	NULL
2440	India	Austr...	Mohali	2006-10-29	Team2	25001	Dhoni	MS	India	1981	Ranchi	2005
2440	India	Austr...	Mohali	2006-10-29	Team2	23001	Singh	Yuvraj	India	1981	Chandigarh	2003
2440	India	Austr...	Mohali	2006-10-29	Team2	21001	Sehwag	Virender	India	1978	Delhi	2001
2449	South ...	India	Cape T...	2006-11-26	Team1	98002	Singh	Harbhajan	India	1980	Jalandhar	1998
2449	South ...	India	Cape T...	2006-11-26	Team1	96003	Ganguly	Saurav	India	1972	Calcutta	1996
2449	South ...	India	Cape T...	2006-11-26	Team1	96001	Dravid	Rahul	India	1973	Indore	1996
2449	South ...	India	Cape T...	2006-11-26	Team1	90002	Kumble	Anil	India	1970	Bangalore	1990
2449	South ...	India	Cape T...	2006-11-26	Team1	89001	Tendulkar	Sachin	India	1973	Mumbai	1989
2449	South ...	India	Cape T...	2006-11-26	Team1	27002	Sharma	Ishant	India	1988	Delhi	2007
2449	South ...	India	Cape T...	2006-11-26	Team1	27001	Kumar	Praveen	India	1986	Meerut	NULL
2449	South ...	India	Cape T...	2006-11-26	Team1	25001	Dhoni	MS	India	1981	Ranchi	2005

LAKSHAY_32_D6EC_DBMS_LAB

MatchID	Team1	Team2	Ground	Date	Winner	PlayerID	LName	FName	Country	Yborn	Bplace	FTest
2449	South ...	India	Cape T...	2006-11-26	Team1	23001	Singh	Yuvraj	India	1981	Chandigarh	2003
2449	South ...	India	Cape T...	2006-11-26	Team1	21001	Sehwag	Virender	India	1978	Delhi	2001
2480	India	West ...	Nagpur	2007-01-21	Team1	98002	Singh	Harbhajan	India	1980	Jalandhar	1998
2480	India	West ...	Nagpur	2007-01-21	Team1	96003	Ganguly	Saurav	India	1972	Calcutta	1996
2480	India	West ...	Nagpur	2007-01-21	Team1	96001	Dravid	Rahul	India	1973	Indore	1996
2480	India	West ...	Nagpur	2007-01-21	Team1	90002	Kumble	Anil	India	1970	Bangalore	1990
2480	India	West ...	Nagpur	2007-01-21	Team1	89001	Tendulkar	Sachin	India	1973	Mumbai	1989
2480	India	West ...	Nagpur	2007-01-21	Team1	27002	Sharma	Ishant	India	1988	Delhi	2007
2480	India	West ...	Nagpur	2007-01-21	Team1	27001	Kumar	Praveen	India	1986	Meerut	NULL
2480	India	West ...	Nagpur	2007-01-21	Team1	25001	Dhoni	MS	India	1981	Ranchi	2005
2480	India	West ...	Nagpur	2007-01-21	Team1	23001	Singh	Yuvraj	India	1981	Chandigarh	2003
2480	India	West ...	Nagpur	2007-01-21	Team1	21001	Sehwag	Virender	India	1978	Delhi	2001
2493	India	west ...	Vadodara	2007-01-31	Team1	98002	Singh	Harbhajan	India	1980	Jalandhar	1998
2493	India	west ...	Vadodara	2007-01-31	Team1	96003	Ganguly	Saurav	India	1972	Calcutta	1996
2493	India	west ...	Vadodara	2007-01-31	Team1	96001	Dravid	Rahul	India	1973	Indore	1996
2493	India	west ...	Vadodara	2007-01-31	Team1	90002	Kumble	Anil	India	1970	Bangalore	1990
2493	India	west ...	Vadodara	2007-01-31	Team1	89001	Tendulkar	Sachin	India	1973	Mumbai	1989

MatchID	Team1	Team2	Ground	Date	Winner	PlayerID	LName	FName	Country	Yborn	Bplace	FTest
2493	India	west ...	Vadodara	2007-01-31	Team1	27002	Sharma	Ishant	India	1988	Delhi	2007
2493	India	west ...	Vadodara	2007-01-31	Team1	27001	Kumar	Praveen	India	1986	Meerut	NULL
2493	India	west ...	Vadodara	2007-01-31	Team1	25001	Dhoni	MS	India	1981	Ranchi	2005
2493	India	west ...	Vadodara	2007-01-31	Team1	23001	Singh	Yuvraj	India	1981	Chandigarh	2003
2493	India	west ...	Vadodara	2007-01-31	Team1	21001	Sehwag	Virender	India	1978	Delhi	2001
2520	India	Sri La...	Rajkot	2007-02-11	Team2	98002	Singh	Harbhajan	India	1980	Jalandhar	1998
2520	India	Sri La...	Rajkot	2007-02-11	Team2	96003	Ganguly	Saurav	India	1972	Calcutta	1996
2520	India	Sri La...	Rajkot	2007-02-11	Team2	96001	Dravid	Rahul	India	1973	Indore	1996
2520	India	Sri La...	Rajkot	2007-02-11	Team2	90002	Kumble	Anil	India	1970	Bangalore	1990
2520	India	Sri La...	Rajkot	2007-02-11	Team2	89001	Tendulkar	Sachin	India	1973	Mumbai	1989
2520	India	Sri La...	Rajkot	2007-02-11	Team2	27002	Sharma	Ishant	India	1988	Delhi	2007
2520	India	Sri La...	Rajkot	2007-02-11	Team2	27001	Kumar	Praveen	India	1986	Meerut	NULL
2520	India	Sri La...	Rajkot	2007-02-11	Team2	25001	Dhoni	MS	India	1981	Ranchi	2005
2520	India	Sri La...	Rajkot	2007-02-11	Team2	23001	Singh	Yuvraj	India	1981	Chandigarh	2003
2520	India	Sri La...	Rajkot	2007-02-11	Team2	21001	Sehwag	Virender	India	1978	Delhi	2001
2611	England	India	Southa...	2007-08-21	Team1	98002	Singh	Harbhajan	India	1980	Jalandhar	1998
2611	England	India	Southa...	2007-08-21	Team1	96003	Ganguly	Saurav	India	1972	Calcutta	1996

MatchID	Team1	Team2	Ground	Date	Winner	PlayerID	LName	FName	Country	Yborn	Bplace	FTest
2611	England	India	Southa...	2007-08-21	Team1	96001	Dravid	Rahul	India	1973	Indore	1996
2611	England	India	Southa...	2007-08-21	Team1	90002	Kumble	Anil	India	1970	Bangalore	1990
2611	England	India	Southa...	2007-08-21	Team1	89001	Tendulkar	Sachin	India	1973	Mumbai	1989
2611	England	India	Southa...	2007-08-21	Team1	27002	Sharma	Ishant	India	1988	Delhi	2007
2611	England	India	Southa...	2007-08-21	Team1	27001	Kumar	Praveen	India	1986	Meerut	NULL
2611	England	India	Southa...	2007-08-21	Team1	25001	Dhoni	MS	India	1981	Ranchi	2005
2611	England	India	Southa...	2007-08-21	Team1	23001	Singh	Yuvraj	India	1981	Chandigarh	2003
2611	England	India	Southa...	2007-08-21	Team1	21001	Sehwag	Virender	India	1978	Delhi	2001
2632	India	Austr...	Mumbai	2007-10-17	Team1	98002	Singh	Harbhajan	India	1980	Jalandhar	1998
2632	India	Austr...	Mumbai	2007-10-17	Team1	96003	Ganguly	Saurav	India	1972	Calcutta	1996
2632	India	Austr...	Mumbai	2007-10-17	Team1	96001	Dravid	Rahul	India	1973	Indore	1996
2632	India	Austr...	Mumbai	2007-10-17	Team1	90002	Kumble	Anil	India	1970	Bangalore	1990
2632	India	Austr...	Mumbai	2007-10-17	Team1	89001	Tendulkar	Sachin	India	1973	Mumbai	1989
2632	India	Austr...	Mumbai	2007-10-17	Team1	27002	Sharma	Ishant	India	1988	Delhi	2007
2632	India	Austr...	Mumbai	2007-10-17	Team1	27001	Kumar	Praveen	India	1986	Meerut	NULL
2632	India	Austr...	Mumbai	2007-10-17	Team1	25001	Dhoni	MS	India	1981	Ranchi	2005
2632	India	Austr...	Mumbai	2007-10-17	Team1	23001	Singh	Yuvraj	India	1981	Chandigarh	2003

LAKSHAY_32_D6EC_DBMS_LAB

MatchID	Team1	Team2	Ground	Date	Winner	PlayerID	LName	FName	Country	Yborn	Bplace	FTest
2632	India	Austr...	Mumbai	2007-10-17	Team1	21001	Sehwag	Virender	India	1978	Delhi	2001
2643	India	Pakistan	Guwahati	2007-11-05	Team1	98002	Singh	Harbhajan	India	1980	Jalandhar	1998
2643	India	Pakistan	Guwahati	2007-11-05	Team1	96003	Ganguly	Saurav	India	1972	Calcutta	1996
2643	India	Pakistan	Guwahati	2007-11-05	Team1	96001	Dravid	Rahul	India	1973	Indore	1996
2643	India	Pakistan	Guwahati	2007-11-05	Team1	90002	Kumble	Anil	India	1970	Bangalore	1990
2643	India	Pakistan	Guwahati	2007-11-05	Team1	89001	Tendulkar	Sachin	India	1973	Mumbai	1989
2643	India	Pakistan	Guwahati	2007-11-05	Team1	27002	Sharma	Ishant	India	1988	Delhi	2007
2643	India	Pakistan	Guwahati	2007-11-05	Team1	27001	Kumar	Praveen	India	1986	Meerut	NULL
2643	India	Pakistan	Guwahati	2007-11-05	Team1	25001	Dhoni	MS	India	1981	Ranchi	2005
2643	India	Pakistan	Guwahati	2007-11-05	Team1	23001	Singh	Yuvraj	India	1981	Chandigarh	2003
2643	India	Pakistan	Guwahati	2007-11-05	Team1	21001	Sehwag	Virender	India	1978	Delhi	2001
2675	Australia	India	Melbourne	2008-02-10	Team2	98002	Singh	Harbhajan	India	1980	Jalandhar	1998
2675	Australia	India	Melbourne	2008-02-10	Team2	96003	Ganguly	Saurav	India	1972	Calcutta	1996
2675	Australia	India	Melbourne	2008-02-10	Team2	96001	Dravid	Rahul	India	1973	Indore	1996
2675	Australia	India	Melbourne	2008-02-10	Team2	90002	Kumble	Anil	India	1970	Bangalore	1990
2675	Australia	India	Melbourne	2008-02-10	Team2	89001	Tendulkar	Sachin	India	1973	Mumbai	1989
2675	Australia	India	Melbourne	2008-02-10	Team2	27002	Sharma	Ishant	India	1988	Delhi	2007

MatchID	Team1	Team2	Ground	Date	Winner	PlayerID	LName	FName	Country	Yborn	Bplace	FTest
2675	Australia	India	Melbourne	2008-02-10	Team2	27001	Kumar	Praveen	India	1986	Meerut	NULL
2675	Australia	India	Melbourne	2008-02-10	Team2	25001	Dhoni	MS	India	1981	Ranchi	2005
2675	Australia	India	Melbourne	2008-02-10	Team2	23001	Singh	Yuvraj	India	1981	Chandigarh	2003
2675	Australia	India	Melbourne	2008-02-10	Team2	21001	Sehwag	Virender	India	1978	Delhi	2001
2681	India	Sri La...	Adelaide	2008-02-19	Team1	98002	Singh	Harbhajan	India	1980	Jalandhar	1998
2681	India	Sri La...	Adelaide	2008-02-19	Team1	96003	Ganguly	Saurav	India	1972	Calcutta	1996
2681	India	Sri La...	Adelaide	2008-02-19	Team1	96001	Dravid	Rahul	India	1973	Indore	1996
2681	India	Sri La...	Adelaide	2008-02-19	Team1	90002	Kumble	Anil	India	1970	Bangalore	1990
2681	India	Sri La...	Adelaide	2008-02-19	Team1	89001	Tendulkar	Sachin	India	1973	Mumbai	1989
2681	India	Sri La...	Adelaide	2008-02-19	Team1	27002	Sharma	Ishant	India	1988	Delhi	2007
2681	India	Sri La...	Adelaide	2008-02-19	Team1	27001	Kumar	Praveen	India	1986	Meerut	NULL
2681	India	Sri La...	Adelaide	2008-02-19	Team1	25001	Dhoni	MS	India	1981	Ranchi	2005
2681	India	Sri La...	Adelaide	2008-02-19	Team1	23001	Singh	Yuvraj	India	1981	Chandigarh	2003
2681	India	Sri La...	Adelaide	2008-02-19	Team1	21001	Sehwag	Virender	India	1978	Delhi	2001
2688	Australia	India	Sydney	2008-03-02	Team2	98002	Singh	Harbhajan	India	1980	Jalandhar	1998
2688	Australia	India	Sydney	2008-03-02	Team2	96003	Ganguly	Saurav	India	1972	Calcutta	1996
2688	Australia	India	Sydney	2008-03-02	Team2	96001	Dravid	Rahul	India	1973	Indore	1996

MatchID	Team1	Team2	Ground	Date	Winner	PlayerID	LName	FName	Country	Yborn	Bplace	FTest
2688	Australia	India	Sydney	2008-03-02	Team2	90002	Kumble	Anil	India	1970	Bangalore	1990
2688	Australia	India	Sydney	2008-03-02	Team2	89001	Tendulkar	Sachin	India	1973	Mumbai	1989
2688	Australia	India	Sydney	2008-03-02	Team2	27002	Sharma	Ishant	India	1988	Delhi	2007
2688	Australia	India	Sydney	2008-03-02	Team2	27001	Kumar	Praveen	India	1986	Meerut	NULL
2688	Australia	India	Sydney	2008-03-02	Team2	25001	Dhoni	MS	India	1981	Ranchi	2005
2688	Australia	India	Sydney	2008-03-02	Team2	23001	Singh	Yuvraj	India	1981	Chandigarh	2003
2688	Australia	India	Sydney	2008-03-02	Team2	21001	Sehwag	Virender	India	1978	Delhi	2001
2689	Australia	India	Brisbane	2008-03-04	Team2	98002	Singh	Harbhajan	India	1980	Jalandhar	1998
2689	Australia	India	Brisbane	2008-03-04	Team2	96003	Ganguly	Saurav	India	1972	Calcutta	1996
2689	Australia	India	Brisbane	2008-03-04	Team2	96001	Dravid	Rahul	India	1973	Indore	1996
2689	Australia	India	Brisbane	2008-03-04	Team2	90002	Kumble	Anil	India	1970	Bangalore	1990
2689	Australia	India	Brisbane	2008-03-04	Team2	89001	Tendulkar	Sachin	India	1973	Mumbai	1989
2689	Australia	India	Brisbane	2008-03-04	Team2	27002	Sharma	Ishant	India	1988	Delhi	2007
2689	Australia	India	Brisbane	2008-03-04	Team2	27001	Kumar	Praveen	India	1986	Meerut	NULL
2689	Australia	India	Brisbane	2008-03-04	Team2	25001	Dhoni	MS	India	1981	Ranchi	2005
2689	Australia	India	Brisbane	2008-03-04	Team2	23001	Singh	Yuvraj	India	1981	Chandigarh	2003
2689	Australia	India	Brisbane	2008-03-04	Team2	21001	Sehwag	Virender	India	1978	Delhi	2001

MatchID	Team1	Team2	Ground	Date	Winner	PlayerID	LName	FName	Country	Yborn	Bplace	FTest
2717	Pakistan	India	Karachi	2008-06-26	Team2	98002	Singh	Harbhajan	India	1980	Jalandhar	1998
2717	Pakistan	India	Karachi	2008-06-26	Team2	96003	Ganguly	Saurav	India	1972	Calcutta	1996
2717	Pakistan	India	Karachi	2008-06-26	Team2	96001	Dravid	Rahul	India	1973	Indore	1996
2717	Pakistan	India	Karachi	2008-06-26	Team2	90002	Kumble	Anil	India	1970	Bangalore	1990
2717	Pakistan	India	Karachi	2008-06-26	Team2	89001	Tendulkar	Sachin	India	1973	Mumbai	1989
2717	Pakistan	India	Karachi	2008-06-26	Team2	27002	Sharma	Ishant	India	1988	Delhi	2007
2717	Pakistan	India	Karachi	2008-06-26	Team2	27001	Kumar	Praveen	India	1986	Meerut	NULL
2717	Pakistan	India	Karachi	2008-06-26	Team2	25001	Dhoni	MS	India	1981	Ranchi	2005
2717	Pakistan	India	Karachi	2008-06-26	Team2	23001	Singh	Yuvraj	India	1981	Chandigarh	2003
2717	Pakistan	India	Karachi	2008-06-26	Team2	21001	Sehwag	Virender	India	1978	Delhi	2001
2750	Sri Lanka	India	Colombo	2008-08-24	Team2	98002	Singh	Harbhajan	India	1980	Jalandhar	1998
2750	Sri Lanka	India	Colombo	2008-08-24	Team2	96003	Ganguly	Saurav	India	1972	Calcutta	1996
2750	Sri Lanka	India	Colombo	2008-08-24	Team2	96001	Dravid	Rahul	India	1973	Indore	1996
2750	Sri Lanka	India	Colombo	2008-08-24	Team2	90002	Kumble	Anil	India	1970	Bangalore	1990
2750	Sri Lanka	India	Colombo	2008-08-24	Team2	89001	Tendulkar	Sachin	India	1973	Mumbai	1989
2750	Sri Lanka	India	Colombo	2008-08-24	Team2	27002	Sharma	Ishant	India	1988	Delhi	2007
2750	Sri Lanka	India	Colombo	2008-08-24	Team2	27001	Kumar	Praveen	India	1986	Meerut	NULL
2750	Sri Lanka	India	Colombo	2008-08-24	Team2	25001	Dhoni	MS	India	1981	Ranchi	2005
2750	Sri Lanka	India	Colombo	2008-08-24	Team2	23001	Singh	Yuvraj	India	1981	Chandigarh	2003
2750	Sri Lanka	India	Colombo	2008-08-24	Team2	21001	Sehwag	Virender	India	1978	Delhi	2001
2755	Sri Lanka	India	Colombo	2008-08-24	Team2	98002	Singh	Harbhajan	India	1980	Jalandhar	1998
2755	Sri Lanka	India	Colombo	2008-08-24	Team2	96003	Ganguly	Saurav	India	1972	Calcutta	1996
2755	Sri Lanka	India	Colombo	2008-08-24	Team2	96001	Dravid	Rahul	India	1973	Indore	1996
2755	Sri Lanka	India	Colombo	2008-08-24	Team2	90002	Kumble	Anil	India	1970	Bangalore	1990
2755	Sri Lanka	India	Colombo	2008-08-24	Team2	89001	Tendulkar	Sachin	India	1973	Mumbai	1989
2755	Sri Lanka	India	Colombo	2008-08-24	Team2	27002	Sharma	Ishant	India	1988	Delhi	2007
2755	Sri Lanka	India	Colombo	2008-08-24	Team2	27001	Kumar	Praveen	India	1986	Meerut	NULL
2755	Sri Lanka	India	Colombo	2008-08-24	Team2	25001	Dhoni	MS	India	1981	Ranchi	2005
2755	Sri Lanka	India	Colombo	2008-08-24	Team2	23001	Singh	Yuvraj	India	1981	Chandigarh	2003
2755	Sri Lanka	India	Colombo	2008-08-24	Team2	21001	Sehwag	Virender	India	1978	Delhi	2001

11 [Natural Join vs Explicit]: Write an SQL query joining Batting and ODIMatch using MatchID to display total runs scored by each player in matches played at 'Colombo' ground.

select o.MatchID, b.PID, o.Ground, b.NRuns from odimatch o join batting b on o.MatchID=b.MatchID where o.Ground='Colombo';

MatchID	PID	Ground	NRuns
2755	23001	Colombo	0
2755	25001	Colombo	71
2755	91001	Colombo	60
2755	94002	Colombo	17
2755	92002	Colombo	1
2755	27001	Colombo	2
2755	98002	Colombo	2

12 [Join with Aggregation]: Display PlayerID, FName, LName, and total career runs (SUM(NRuns)) across all matches by joining Player and Batting, sorted in descending order of total runs.

select b.PID, p.FName,p.LName,sum(NRuns) as "Tot_Carr_runs" from player p join batting b on p.PlayerID=b.PID group by b.PID order by Tot_Carr_runs DESC;

PID	FName	LName	Tot_Carr_runs
25001	MS	Dhoni	107
89001	Sachin	Tendulkar	91
91001	Sanath	Jayasuriya	60
24001	Andrew	Symonds	42
23001	Yuvraj	Singh	38
94002	Chaminda	Vaas	17
27001	Praveen	Kumar	9
99001	Breet	Lee	7
98002	Harbhajan	Singh	5
99002	Adam	Gilchrist	2
92002	Muthiah	Murlitharan	1
95001	Ricky	Ponting	1

13 [Left Join Anti-Join]: Write an SQL query using LEFT JOIN to identify all players (PlayerID, FName, LName, Country) who have NEVER recorded a bowling performance in the Bowling table.

select p.PlayerID, p.FName, p.LName, p.Country from player p where not exists(select bow.PID from bowling bow where p.PlayerID = bow.PID);

PlayerID	FName	LName	Country
21001	Virender	Sehwag	India
25001	MS	Dhoni	India
27001	Praveen	Kumar	India
27002	Ishant	Sharma	India
89001	Sachin	Tendulkar	India
90001	Brain	Lara	West Indies
90002	Anil	Kumble	India
92001	Shane	Warne	Australia
92003	Inzamam	UI-Huq	Pakistan
93002	Heath	Streak	Zimbabwe
93003	Gary	Kirsten	South Africa
94004	Stephen	Fleming	New Zealand
95001	Ricky	Ponting	Australia
95002	Shaun	Pollock	South Africa
95003	Jacques	Kallis	South Africa
96001	Rahul	Dravid	India
96002	Herschelle	Gibbs	South Africa
96003	Saurav	Ganguly	India
97004	Daniel	Vettori	New Zealand
98001	Shahid	Afridi	Pakistan
99002	Adam	Gilchrist	Australia
99003	Michael	Vaughan	England
300311	Michael	NULL	NULL
NULL	NULL	NULL	NULL